PROBLEM → RESULT

Where is the race car? *Cameras only*, at race speed, with no start pose.

Causal x, y, yaw at 10 Hz from the car's cameras, wheel speeds and steering. No GNSS, no LiDAR, no starting pose given. A geo-referenced visual map, odometry and an EKF, all classical and explainable.

POSITION RMSE
0.51 m

LATERAL RMSE
0.30 m

HEADING RMSE
0.31 °

POSES WITHIN 1 M · SCORED METRIC
95.1 %

1-S INTERVALS WITH A POSE · 90 % NEEDED
99.8 %

TIME PER FRAME · M4 CPU · 100 MS BUDGET
24–39 ms

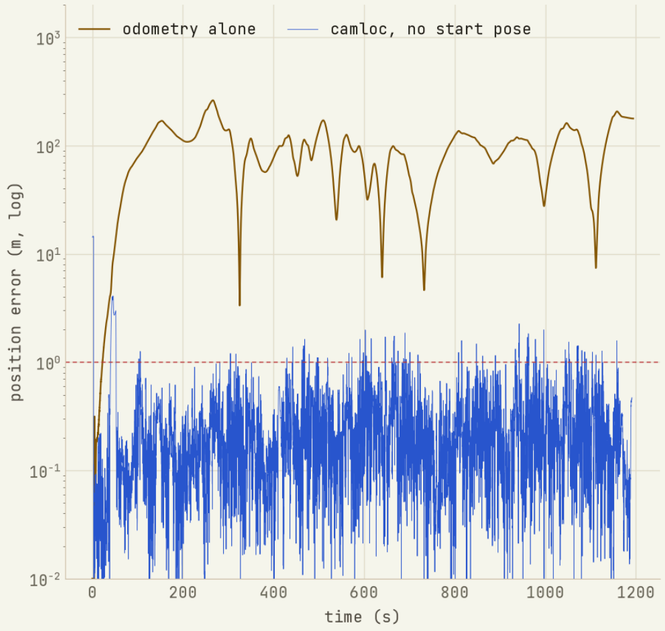
CLASSICAL · EXPLAINABLE | CAUSAL | CPU ONLY | NO PRE-TRAINED MODELS | NO EXTERNAL DATA | NO START POSE

Scope: held-out night laps 2 / 4 / 6 of the training run, pooled; the visual map and every calibration exclude the scored lap; no start pose. Coverage: full training run via run.sh (no pose during the first 3 s of search). Runtime: mean per frame, full run (24 ms) and 4 parallel folds (39 ms). Not a hidden-test result.

WHY NOT JUST ODOMETRY?

Wheels and steering alone drift *hundreds of metres*.



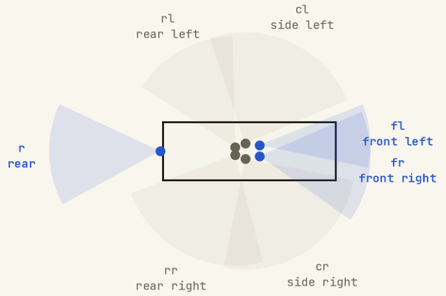


ODOMETRY ALONE · WHOLE RUN

107 m RMSE

up to 263 m off, even starting from the true pose

7 CAMERAS · 10 HZ · NIGHT



fl · fr · r carry the map fixes; the side cameras sit about 0.2 m above the paint, so lateral position from lines is unobservable.

Scope: training run, 1195 s, night. Odometry: calibrated wheel speeds + steering (bicycle model), started from the GT pose (display only). Ground truth is drawn for display only. Camera images are not shown (dataset licence: no redistribution).

THE GUIDING ANALOGY

A rally *co-driver* with a pace-note book.

THE CO-DRIVER…	…IS THIS BLOCK IN CAMLOC (CAUSAL, EVERY FRAME)
Counts distance and turns from the car's own speed and steering between landmarks. Smooth, but it drifts.	ODOMETRY + RELATIVE YAW Wheel speeds + steering with a braking-aware axle choice (or validation/twist); keyframe rotation from LK flow + essential matrix.
Recognises a landmark from the pace notes and snaps back to the map. Absolute, but only now and then.	GEO-REFERENCED MAP + PNP CLAHE ORB features matched to a 3D map built from training laps; PnP + RANSAC per camera gives an absolute fix in metres.
Judges each sighting: do two eyes agree, and does it fit where we should be? Look-alikes are rejected.	CONSENSUS + NIS GATE + EKF Cross-camera agreement, a statistical consistency test that can recover, and an EKF (x, y, yaw, sideslip) on track and pit geometry.
Dropped mid-stage with no position: looks around until sightings agree, commits provisionally, and changes their mind if the evidence says so.	NO START POSE Global search over the map → provisional lock (2 cameras or 3 frames agree) → confirm after 2 accepted fixes, or switch when ≥ 3 rejected fixes agree elsewhere.

Declarations: no pre-trained models, no external data; OpenCV ORB / LK / PnP only. Car signals: wheel speeds, steering, brake (allowed set, ADR-0003). The GT pose and /tf map→base_link are never read at inference.

WHY EACH CHOICE · AND THE EVIDENCE

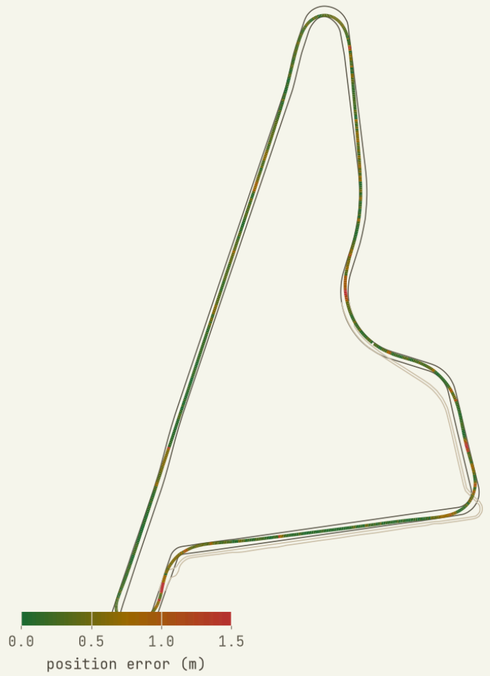
Every block earns its place *with a measurement*.

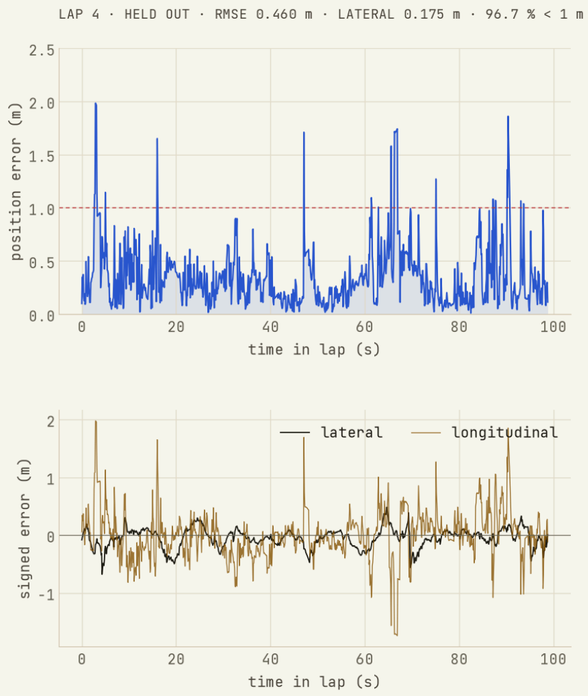
CHOICE	WHY	EVIDENCE
Classical, not deep learning or VLMs	One night of data, so nothing to overfit; explainable; deterministic; real-time on a CPU	`24-39 ms` per frame on an M4, against a 100 ms frame budget
Geo-referenced map + PnP	Known 3D geometry gives absolute metres, not just relative motion	Map fixes take the error from `107 m` (odometry alone) to under 1 m
Odometry + EKF	Smooth, causal, and it fills the gaps between fixes	30 s without any map fix: `2.7 m`, and it recovers when fixes return
Gating + consensus	A wrong landmark is worse than none: reject outliers before they reach the filter	2 % of fixes injected 10 m wrong: lap 4 `0.457 m` vs 0.460 m clean
Provisional lock	Commit early so the output is never silent, but never trust a single sighting	`16 / 16` mid-lap starts end correct; a 13.6 m alias switches in 0.3 s
Strict validation	The reason these numbers can be trusted: map and every calibration rebuilt without the scored lap; the official scorer; causal replay	Same `0.513 m` with or without a GT start pose; calibration leakage measured at ~0.02 m

Scope: night training run with held-out laps 2 / 4 / 6 (strict); wrong-fix and blackout tests on lap 4; odometry baseline over the whole run from the GT start pose (display only). Experiment IDs: EXP-0001, 0017, 0019, 0035, 0036-0039.

RESULTS · HELD-OUT LAPS

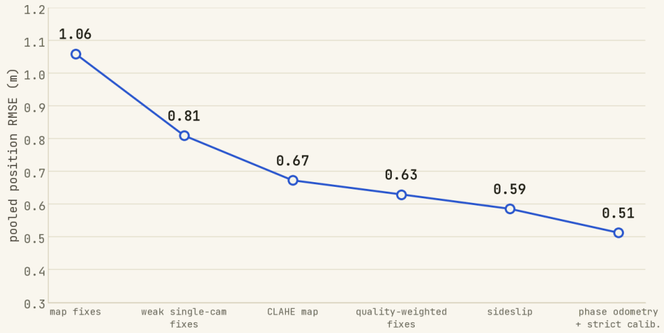
Sub-metre on every held-out lap, *from any start*.





HELD-OUT LAP	POS RMSE	LATERAL
Lap 2	0.598 m	0.404 m
Lap 4 (fastest)	0.460 m	0.175 m
Lap 6	0.415 m	0.206 m
Pooled	**0.513 m**	**0.303 m**

POOLED RMSE BY DEVELOPMENT STEP



16 MID-LAP STARTS, NO START POSE

16 / 16 end correct. Lock in 0.0–1.8 s. An aliased lock 13.6 m off **switches after 0.3 s**; one 9.9 m off converges by 11 s.

Scope: night training run, laps 2 / 4 / 6 each held out: the map and every calibration are rebuilt without the scored lap (strict). The progression points before the last one exclude only the scored lap from the map. Mid-lap starts: 40 s windows at 10–90 % of each held-out lap.

FAILURE CASES & LESSONS LEARNED

We know exactly *where it degrades*, and by how much.

MEASURED LIMITS

30 s / 60 s without map fixes	2.7 m	5.2 m
Aliased start: time to correct	0.3–11 s	
Daylight sample: accepted fixes	11 in 74 s	
Lateral from paint (0.2 m height)	unobservable	
Front exposure (blur small to ~60 m/s)	0.5–1 ms	

LESSONS LEARNED

1. **Strict hold-out is what makes the numbers trustworthy:** map and every calibration rebuilt without the scored lap (leakage ~0.02 m).
2. **Check lock rules on test-domain data:** a pair-required lock passed every night test but never locked on the daylight sample; the fix was a provisional lock plus confirmation.
3. **Map fixes, not odometry, decide accuracy:** 107 m alone, sub-metre with fixes.

```
$ ./run.sh test.mcap out.csv
```

No flags. Both topic layouts. Builds the venv and the visual map on first use.

- Strict leave-one-lap-out
- Causal, image stamps only
- GT pose never read
- 27 automated tests
- Python 3.10–3.12, CPU
- 39 logged experiments

GITHUB.COM/MULAYDM10/YAS-CAMLOC

TEAM

Rogue *Ravens*

Tamo Schäfers · Dhruv Mulay · Gagan Karthik Sripuram · Taa-seen Ahmed

Scope: blackouts on lap 4; blur at 60 m/s extrapolated. Night training run with held-out laps, except the organisers' daylight sample (structural check, no ground truth, never used for tuning). run.sh with no flags was checked on the full training run and on a 60 s test-domain cut.